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8 MAY 1963
SUPERSEDING
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MILITARY SPECIFICATION

FIRING KIT, DEMOLITION, ELECTRICAL XM-147E1

1. SCOPE

1.1 This specification covers the quality assurance provisions, and requirements not covered by drawings, for one type of firing mechanism designated as Firing Kit, Demolition, Electrical, XM-147E1. (see 6.4 and 6.5)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 — Ammunition and Special Weapons; General Specification for.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-109 — Inspection Terms and Definitions.

MIL-STD-1235 — Simple and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

ORDNANCE CORPS

8865497 — Firing Kit, Demolition, Electrical, XM-147E1.

PUBLICATIONS

ORDNANCE CORPS

IEL-8865497 — Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings and publications required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Material.** Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 **Assemblies.** The assemblies shall comply with all requirements specified on Drawing (dwg.) 8865497 and with all requirements specified in applicable specifications.

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3.3 Mechanical functioning of firing switch. It shall not be possible to change from one selector position to another without first moving the lever in the direction opposite to the new position, it shall not be possible to skip a position and it shall be possible to reach all positions. Testing shall be as specified in 4.3.1.

3.4 Electrical functioning of firing switch. The switch shall not close any circuit when in the "off" position and shall close the appropriate circuit only when in the other positions. When the switch is in the "power" position the light shall be lit. With Cable Nos. 1 and the 3 conductor cable assembled to the firing switch the resistance of the rocket and fuze circuits shall not exceed 2 ohms. With Cable No. 2 replacing Cable No. 1 neither circuit shall exceed 6 ohms. Testing shall be as specified in 4.3.1.

3.5 Leakage test. When the case lid and cover are assembled one to the other, there shall be no evidence that leakage may occur. Testing shall be as specified in 4.3.2.

3.6 Welds. The welds shall be reasonably smooth and free of craters, porosity or cracks. They shall exhibit characteristics of fusion, penetration and soundness of weld deposit representative of good welding practice. They shall be free of irregularities which indicate lack of skill or experience on the part of the operator. All scale and weld splatter shall be removed prior to application of protective coatings.

3.7 Workmanship. All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, unblended radii, surface defects, chips, dirt, grease, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Surface coatings shall be continuous except for a few slight scratches not exposing base material. All required marking shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. The supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 in order to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 *Contractor quality assurance system.* The contractor shall provide and maintain an effective quality assurance system in compliance with the requirements of MIL-I-45208.

4.1.2 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.

(b) Number of units of product inspected.

(c) Results obtained for all inspection performed.

(d) Drawing, specification number and date, together with an identification and date of changes.

(e) Certificates of analysis on all material purchased by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.

(f) Number of items in the lot.

(g) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his prin-

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cipal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 *Government verification.* Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be in accordance with MIL-I-45208 and will be performed to the exent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Inspection provisions.

4.2.1 *Lot formation.* A lot shall consist of firing kits produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification, and process changes not affecting safety, performance or fit, as determined by the Government, shall not necessitate changing the lot interfix number.

4.2.2 *Examination.* Sampling plans and procedures for the following classifications of defects shall be in accordance with MIL-STD-105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Cables No. 1, No. 2 and No. 3 conductor (see dwg. 8865475, 8865476, or 8865477 covering details of dwg. 8865497).

Categories	Defects	Method of inspection	Code No. (see 6.3)
Critical:	None defined.		
Major:	AQL 0.25 percent		
101.	Cable or clamp damaged	Visual	01001
Minor:	AQL 1.50 percent		
201.	Length	Gage	01002
202.	Connector clamp loose	Manual	01003
203.	Evidence of poor workmanship (see 3.7)	Visual	01004

4.2.2.2 *Cover, prior to paint* (see dwg. 8865474 covering a detail of dwg. 8865497).

Categories	Defects	Method of inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Reel action not free	Manual	02001
102.	Component loose	Manual	02002
103.	Component missing (check list)	Visual	02003
104.	Weld inadequate (see 3.6)	Visual	02004
Minor:	AQL 1.00 percent		
201.	Protective coating missing or inadequate	Visual	02005
202.	Evidence of poor workmanship (see 3.7)	Visual	02006

4.2.2.3 *Lid, case, prior to paint* (see dwg. 8865454 covering a detail of dwg. 8865497).

Categories	Defects	Method of inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Component missing	Visual	03001
102.	Weld inadequate	Visual	03002
103.	Component loose	Manual	03003
Minor:	AQL 1.00 percent		
201.	Protective coating missing or inadequate	Visual	03004
202.	Evidence of poor workmanship (see 3.7)	Visual	03005

4.2.2.4 *Assembly, prior to closing* (see dwg. 8865497, 2 of 2).

Categories	Defects	Method of inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.25 percent		
101.	Component missing or loose (check list)	Visual-Manual	04001
102.	Reel action not free	Manual	04002

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Categories	Defects	Method of inspection	Code No.
Minor:	AQL 1.00 percent		
201.	Protective coating missing or inadequate	Visual	04003
202.	Evidence of poor workmanship (see 3.7)	Visual	04004

4.2.2.5 *Assembly, prior to closing* (see dwg. 8865497, 1 of 2).

Categories	Defects	Method of inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.25 percent		
101.	Content unit missing	Visual-Manual	05001
Minor:	AQL 0.40 percent		
201.	Evidence of poor workmanship (see 3.7)	Visual	05002

4.2.2.6 *Assembly, strapped* (see 5.1.1).

Categories	Defects	Method of inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.25 percent		
101.	Strap loose	Manual	06001
102.	Corner protector missing	Visual	06002
Minor:	AQL 0.40 percent		
201.	Marking missing, misleading or unidentifiable	Visual	06003

4.2.3 *Testing.*

4.2.3.1 *Mechanical functioning of firing switch* (see 3.3). Major defect—Code No. 07001. This test shall be performed 100 percent as specified in 4.3.1. Failure in meeting one or more of the requirements shall class a unit defect and it shall be removed from the lot subject to repair and resubmission to test.

4.2.3.2 *Electrical functioning of firing switch* (see 3.4). Major defect—Code No. 08001. This test shall be performed 100 percent as specified in 4.3.1 using the applicable test equipment of 4.2.4. Failure in meeting one or more of the requirements shall class a subassembly defective and it shall be removed from lot subject to repair and resubmission to test.

4.2.3.3 *Leakage test* (see 3.5). Major defect—Code No. 05004. This test shall be performed 100 percent as specified in 4.3.2. Any unit failing to meet the requirement shall be classed defective and removed from the lot subject to rework and resubmission to test.

4.2.4 *Inspection equipment.* Index of Inspection Equipment Number IEL-8865497 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.2.4.1 *Government rights to documentation.* Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.2.4.2 *Supply and maintenance.* Supply and maintenance of the equipment listed on the IEL shall be in accordance with MIL-I-45607.

4.3 Test methods and procedures.

4.3.1 *Mechanical and electrical functioning of firing switch.* Cable No. 1 and the 3 conductor cable (assembled on the reel) shall be connected to the switch. A 24-volt direct current source shall be connected between the clips of the No. 1 cable and the switch moved to the POWER position. Observation shall be made for lighting of the light. The power source shall be disassembled and the test equipment connected between the leads. Terminals A and B of the 3 conductor connector shall be shorted, the switch shall be moved to the ROCKET position and the resistance observed. The short shall be removed and connected to the B and C terminals. The switch shall be moved to the FUZE position and the resistance observed. The switch shall be returned to the "OFF" position, the No. 1 cable replaced by the No. 2 cable and the test repeated. Any subassembly failing one or more of the requirements shall be classed defective.

Note. Accepted sets of switch and 3 cables shall be identified as a unit and placed in assemblies as such.

4.3.2 *Leakage test.* Strips of bond stationery 4 inches by one inch shall be placed at each

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corner of the cover covering the seam, 4 strips shall be placed, equally spaced visually, over each side and 3 strips between the closure clasps. Each strip shall be so placed that the end is above the inside edge of the gasket and shall be bent 90 degrees at the edge of the cover. The lid shall be assembled. Each strip shall be pulled with the force applied perpendicular to the side of the assembly. Sufficient force shall be applied to rupture or withdraw the strip. In the event the strip is withdrawn without a very perceptible drag, the unit shall be classed defective.

5. PREPARATION FOR DELIVERY

5.1 Packing

5.1.1 *Level C.* Two 5/8-inch by .020-inch straps shall be secured around the assembly with corner protectors between the straps and each corner. Each strap shall be placed not more than 1 inch inside the closure clasps.

5.2 **Data cards.** Data card information will be prepared for each lot in accordance with the procedure specified in Specification MIL-A-2550.

6. NOTES

6.1 **Ordering data.** Procurement documents shall specify the following:

(a) Title, number and date of this specification.

6.2 **Inspection code numbers.** The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 **Inspection equipment.** The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists in accordance with the instructions of paragraphs 6.3.1 through 6.3.7.

6.3.1 *Inspection equipment lists.* Inspection equipment lists (IEL) indicate the availability of inspection equipment design by showing in the "number" column of the list of inspection equipment (00 Form 1242-3) the numbers of drawings of existing equipment designs or codes indicating "commercial item" or "contractor design". Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.3.2 and 6.3.3. Action required by the contractor with respect to "commercial item" inspection equipment is described in paragraph 6.3.4. The contractor will be required to prepare drawings for all the equipment coded as "contractor design" in the number column. The code will further indicate whether detailed or specification control drawings are required. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 *Ordnance designs.* Ordnance designs are detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Ordnance design is listed for an item of inspection equipment. Ordnance designs fall into two basic classifications; mandatory and nonmandatory. When an IEL references mandatory Ordnance designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with nonmandatory designs and with the approval of the Government, design alternate inspection equipment to facilitate his operations. Such contractor prepared designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.3 *Specification control drawings.* Specification control drawings depict the device in outline, descriptive, diagrammatic, or pictorial form only and specify the required performance or other characteristics. Contractors must prepare drawings of their designs in support of specification control drawings. The IEL code will indicate whether drawings must be detailed. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirement of specification control drawings may be approved if described in suffi-

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cient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 *Commercial items.* Commercial item inspection equipment is equipment that has a universal application for a specific function. It is comprised of items commonly used by industry and government. Contractors are not required to furnish drawings of commercial item inspection equipment but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the agency administering the contract.

6.3.5 *IEL codes.* The inspection equipment referenced in paragraphs 6.3.1, 6.3.2, 6.3.3, and 6.3.4 will be designated in the IEL by the following codes:

CDOF — Contractor's design responsibility on Ordnance format; ORDM-608-12.

CDCF — Contractor design responsibility on contractor format.

OD — Ordnance design.

ODMU — Ordnance design, mandatory for use.

CE — Commercial equipment.

SCD — Specification control drawing.

6.3.6 *Use of standard measuring equipment.* If the contractor desires to use standard measuring equipment in lieu of government or contractor designed gages, diagrammatic sketches or other descriptions of the use of such equipment shall be submitted for approval as specified in 6.3.7.

6.3.7 *Submission of contractor designs.* Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissible. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

6.4 **Government furnished material.** The procuring activity shall furnish a sufficient quantity of Battery, Dry Cell, Type BA-414/U, Specification MIL-B-18, Galvanometer, Type 1, Specification W-B-411 and Knife, Electricians, Federal Stock Number 7340-721-6811 for each procurement.

6.5 **Intended use.** It is intended that the item of this specification will be used in Demolition Kit, Projected Charge, HE, M125 and M126.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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